

## Assignment 3 - Data Structures for a Community Festival

Submission Date: Tuesday, September 22, 2026 (by 11.59pm on Canvas)

This assignment practices the basic data structures introduced in the Week 3 notebook, 3\_01\_Data\_Structures.ipynb. You will use short, simple examples to organize a fictional community festival. Work one part at a time and use the notebook examples and hints for help. Loops, functions, additional packages and copying nested data structures are not required.

Complete the work in a new Google Colab notebook. Put your name, course number, and assignment title at the top; label every part clearly; run every code cell; and leave all output visible. Use descriptive snake\_case names and include units where appropriate.

Table 1 outlines the required assignment components.

| Component                               | Requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Points |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <b>1. Create a festival route</b>       | Create festival_route = ["Entrance", "Duck pond", "Music lawn", "Exit"]. Print the whole list, the first stop and the last stop. Print the number of stops using len().                                                                                                                                                                                                                                                                                                                                                                                               | 10     |
| <b>2. Update your route</b>             | Continue with festival_route from Part 1. Replace "Music lawn" with "Cupcake stall". Add "Picnic meadow" using .append(), then remove "Exit" using .remove(). Print the updated list and its length. Use a slice to print its first two stops.                                                                                                                                                                                                                                                                                                                        | 15     |
| <b>3. Store a coordinate pair</b>       | Create entrance_coordinates = (38.830, -77.310). This tuple stores latitude followed by longitude in decimal degrees. Print the whole tuple, then use indexing to print latitude and longitude separately with labels. Print its length. In one sentence, explain why a tuple is useful for a coordinate pair that should not change. You do not need to modify the tuple.                                                                                                                                                                                            | 10     |
| <b>4. Create a stall dictionary</b>     | Create cupcake_stall = {"name": "Cupcake Corner", "cupcakes": 24, "open": True}. Print the name using its key. Six cupcakes are sold: subtract 6 from the stored cupcakes value and save the result back to that key. Add a new key, "closing_time", with value "16:00". Print the final dictionary.                                                                                                                                                                                                                                                                  | 20     |
| <b>5. Find shared activities</b>        | Create morning_activities = ["music", "crafts", "music"] and afternoon_activities = ["crafts", "picnic"]. Convert each list to a set using set(). Print both sets and compare the length of the morning list with the length of its set. Use .intersection() to find the shared activities and .union() to find all distinct activities. In one sentence, explain why the repeated "music" appears only once in the set. The order of printed set elements does not matter.                                                                                           | 15     |
| <b>6. Clean a location label</b>        | Create raw_name = " Cupcake Corner ". Use .strip() to save a cleaned version as clean_name. Print clean_name, then print clean_name.upper() and clean_name.lower(). Print raw_name again. In one sentence, explain whether the original string changed.                                                                                                                                                                                                                                                                                                               | 10     |
| <b>7. Design your own festival stop</b> | Invent a fictional festival stop. Create one dictionary named my_stop with three keys: "name" (a string), "coordinates" (a tuple containing latitude and longitude in decimal degrees), and "activities" (a list of two activity strings). Print the dictionary, then print each of its three values using the relevant key. Print the number of activities using len(my_stop["activities"]). In two sentences, describe your stop and explain why a list is useful for its activities. Follow the examples from Parts 1, 3 and 4; no list of dictionaries is needed. | 20     |

Table 1 Assignment components and requirements.

Submit the completed .ipynb file online. Before submitting, restart the runtime and run the notebook from top to bottom. Check that all seven parts, requested outputs and short explanations are present,

and no unexpected errors remain. Use Python operations rather than manually typing calculated results. All locations are fictional; coordinate tuples store latitude followed by longitude in decimal degrees.

Also submit a .PDF file of your notebook (or a .PDF of screenshots), for easy review.

Table 2 outlines the assessment criteria. Component points total 100; the criteria describe how the work is assessed across those components.

| Criterion                             | Evidence                                                                                                                                                 | Weight |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <b>Correctness / completeness</b>     | All seven parts are present; basic list operations, coordinate indexing, dictionary updates, set comparisons and string methods produce correct results. | 40%    |
| <b>Data structure selection / use</b> | Lists, tuples, dictionaries, sets and strings are created and used appropriately with the methods practiced in the notebook.                             | 25%    |
| <b>Documentation / readability</b>    | Names preferably follow snake_case, labels and coordinate units are clear, and the requested short explanations are understandable.                      | 15%    |
| <b>Testing / output</b>               | All requested operations are run and their outputs are visible. The final festival stop includes all three required keys and the requested data types.   | 15%    |
| <b>Organization / submission</b>      | The notebook is clearly labeled, logically ordered, complete, and submitted in the required formats.                                                     | 5%     |

Table 2 Assessment criteria.

On this particular assignment, you are not allowed to use AI, otherwise it defeats the point of trying to learn this content. All the operations you need are covered in the Week 3 notebook, and you are allowed to read the Python documentation for extra background. The knowledge from this assignment is cumulative and necessary to succeed in later weeks of the course (and especially in the pencil and paper exam where you cannot use AI). Undisclosed AI-generated work, fabricated material, or code that the student cannot explain may constitute an Honor Code violation.